

Zhuoyun Du

Email: duzy@zju.edu.cn | Phone: (+86) 186-3036-8946
[Homepage](#) | [GitHub](#) | [Google Scholar](#)

RESEARCH INTERESTS

My research asks whether agents can **learn to design their own communication protocols, coordination rules, and improvement mechanisms**, rather than relying on human-specified ones. I pursue this through three connected threads:

1. **Latent-Space Communication for Agentic Foundation Models:** Treating latent representations as a native medium for reasoning and coordination among agents, and as a bridge between agent design and core representation learning.
2. **Self-Improving Multi-Agent Systems:** Building agents that refine coordination and problem-solving through interaction, feedback, and textual parameter optimization, moving beyond fixed workflows.
3. **Autonomous Evaluation & Post-Training:** Exploring agent-driven evaluation, preference alignment, and data-generation loops that let models improve in open-ended settings with less dense human supervision.

EDUCATION

Zhejiang University, Hangzhou, China
M.Eng. in Data Science and Engineering
State Key Lab of CAD & CG

Sep 2024 – Jul 2027 (Expected)
GPA: 3.95 / 5.00
Advisor: [Prof. Wei Chen](#)

Jinan University, Guangzhou, China
B.Eng. in Computer Science and Technology

Sep 2020 – Jul 2024
GPA: 4.00 / 5.00; Rank: 3 / 99

SELECTED PUBLICATIONS

[†] indicates co-first / equal contribution.

First-author / Co-first-author

1. **Zhuoyun Du**[†], Runze Wang[†], Huiyu Bai, Zouying Cao, Xiaoyong Zhu, Bo Zheng, Wei Chen and Haochao Ying. Enabling Agents to Communicate Entirely in Latent Space [**Interlat**]. *ACL 2026*. [\[paper\]](#) [\[Code\]](#)
2. **Zhuoyun Du**[†], Lujie Zheng[†], Renjun Hu, Yuyang Xu, Xiawei Li, Ying Sun, Wei Chen, Jian Wu, Haolei Cai, and Haochao Ying. LLMs can simulate standardized patients via agent coevolution [**EvoPatient**]. *ACL 2025*. [\[paper\]](#) [\[Code\]](#)
3. **Zhuoyun Du**[†], Chen Qian[†], Wei Liu, Zihao Xie, Yifei Wang, Yufan Dang, Weize Chen, and Cheng Yang. Multi-agent software development through cross-team collaboration [**Croto**]. *Findings of ACL 2025*. [\[paper\]](#) [\[Code\]](#)
4. Huiyu Bai[†], Runze Wang[†], **Zhuoyun Du**[†], Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng and Xuejiao Zhao. Online-PVLM: Advancing Personalized VLMs with Online Concept Learning [**Online-PVLM**]. *Submitted to MM 2026*. [\[arXiv\]](#)

Co-authored

1. Shan He, Runze Wang, **Zhuoyun Du**, Huiyu Bai, Zouying Cao, Yu Cheng, Bo Zheng. Learning to Evolve: A Self-Improving Framework for Multi-Agent Systems via Textual Parameter Graph Optimization [**Learning to Evolve**]. *Findings of ACL 2026*. [\[paper\]](#)
2. Yuewen Liu, Peng Xu, **Zhuoyun Du**, Muxi Diao, Anyi Zhang, Yang Li, Yutong Zhang. MemCoRL: Alternating Co-Optimization of Memory Retrieval and Utilization via Collaborative Reinforcement Learning [**MemCoRL**]. *ACL 2026*.
3. Chen Qian[†], Zihao Xie[†], Yifei Wang[†], Wei Liu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, Zhiyuan Liu, and Maosong Sun. Scaling large-language-model-based multi-agent collaboration [**MACNet**]. *ICLR 2025*. [\[paper\]](#)
4. Wei Liu[†], Chenxi Wang[†], Yifei Wang, Zihao Xie, Rennai Qiu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, and Chen Qian. Autonomous agents for collaborative task under information asymmetry [**iAgent**]. *NeurIPS 2024*. [\[paper\]](#)

RESEARCH & INDUSTRY EXPERIENCE

ByteDance, Data Department
Talent Program Intern

Mar 2026 – Present

- Built the team's **first** data-flywheel rollout system for multi-turn, task-oriented dialogue, providing trajectory-generation infrastructure for downstream **multi-turn dialogue RL** and **agentic RL** experiments.
- Designed a **self-play simulation and evaluation environment** with **high-fidelity, persona-conditioned user-simulation agents**, generating multi-turn dialogue rollouts and **evaluation** signals for downstream RL fine-tuning.
- Conducted research on **LLM post-training and preference alignment** (SFT, RLHF/DPO) and **autonomous, agent-driven evaluation** of the resulting dialogue policies.

Alibaba, Future Living Lab / Token Foundry with Tongyi
Research Intern

Jan 2025 – Mar 2026

- Conducted research within a frontier foundation-model ecosystem whose broader initiatives include HappyHorse (#1 on the Artificial Analysis Video Arena) and HappyOyster (real-time interactive world model), focusing on latent-space communication for multi-agent reasoning.
- Proposed **Interlat**, a latent-space communication framework that replaces natural-language message passing between agents with temporally aligned hidden-state exchange; designed a token-latent curriculum and a learned compression scheme, achieving up to a 24× communication speedup over natural-language message passing at comparable task performance; gained hands-on experience with distributed training on 128+ GPUs.
- Co-developed **Online-PVLM**, on online concept learning for personalized VLMs (**submitted to MM 2026**); also co-authored **MemCoRL (ACL 2026)** and **Learning to Evolve (Findings of ACL 2026)**.

Zhejiang University, State Key Lab of CAD & CG

Sep 2024 – Present

M.Eng. Research, advised by Prof. Wei Chen

- Led **EvoPatient (ACL 2025)**, an open-source doctor-patient coevolution system for virtual standardized patients, adopted by several Chinese tertiary-care hospitals for medical-student teaching, residency training, and OSCE-style standardized-patient practice.
- Drove the project end-to-end, including research formulation, hospital-side requirement gathering, coevolution-method design, core system implementation, prompt engineering, experiments, medical-case curation/de-identification, expert evaluation, and paper writing.
- Improved the patient-answer *Ability* score (average of relevance, faithfulness, and robustness) from 0.763 (Few-shot baseline) to 0.860 after 200 training cases. [\[paper\]](#)

Tsinghua University, THUNLP

Dec 2023 – Aug 2024

Research Assistant

- Worked under Prof. **Zhiyuan Liu**'s supervision on **ChatDev**-based multi-agent collaboration, studying how LLM agents organize roles, exchange information, and coordinate under cooperative or asymmetric settings.
- Led **Croto (Findings of ACL 2025)** as a ChatDev-branch project, designing cross-team orchestration with parallel teams, hierarchy partitioning, and greedy aggregation for multi-agent software development.
- Contributed to ChatDev-branch projects including **MACNet (ICLR 2025)** and **iAgent (NeurIPS 2024)**; related OpenBMB/ChatDev open-source projects have accumulated over **33k stars** on GitHub.

SKILLS, HONORS, SERVICE & BACKGROUND

- **LLM & Machine Learning:** Multi-agent system design; distributed training on **128+ GPUs**; SFT, LoRA, RLHF/DPO; DeepSpeed, FlashAttention.
- **Programming Languages & Frameworks:** Python, C/C++, Bash, Git, L^AT_EX; PyTorch, HuggingFace ecosystem (*e.g.*, Transformers, Datasets).
- **Languages:** English (IELTS: 7.5); Mandarin (Native).
- **Honors:**
 - 2025 **National Scholarship**
 - Outstanding Graduate Student Scholarship
 - **3 Undergraduate Scholarships**
 - **First-Class Scholarship for Outstanding Graduates**
- **Service:** Program Committee, NeurIPS 2025 and ICLR 2025.
- **Interests & Background:** Experienced in collaborative research and communication; former **National Champion in Amateur Épée Fencing**; Grade 7 Piano Certificate; interests include snowboarding, photography, and arts and culture.